

1] A reversible process is one where if you go from State 1 to 2 along the same thermodynamic path as State 2 back to 1 both the system and its surroundings return to the original state. In an irreversible process, the surroundings does not return to its original state due to entropy generation and the additional work it takes to return the system to its original state.

Because of this additional entropy generation and work, irreversibilities reduce the thermal efficiency of heat engines. To avoid irreversibilities, reduce friction through lubrication and heat loss through insulation.

2] Given: $T_H = 800 \text{ K}$
 $T_L = 300 \text{ K}$
 $Q_L = -10 \text{ kJ/s}$

Assume: reversible
 steady-state
 quant. of processes

a) reversible $\rightarrow \eta_{th} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{800}$
 $\eta_{th} = 0.625$

b) Carnot cycle: heat addition is isothermal at T_H

$$\Delta S = \oint_{rev} \frac{\delta Q}{T} = \frac{1}{T_H} \int \delta Q = \frac{Q_H}{T_H}$$

$$\Rightarrow \text{Reversible: } \frac{Q_H}{Q_L} = \frac{T_H}{T_L} \rightarrow Q_H = Q_L \frac{T_H}{T_L} = 10 \left(\frac{800}{300} \right) = 26.67 \text{ kW}$$

$$\Rightarrow \Delta S = \frac{26667}{800} \left[\frac{\text{J}}{\text{K}} \right] \rightarrow \Delta S = 33.333 \frac{\text{J}}{\text{K}}$$

c) Carnot cycle: heat rejection is isothermal

$$\Delta S = \oint_{\text{rev}} \frac{\delta Q}{T} = \frac{Q_L}{T_L} = \frac{-10000}{300} \rightarrow \boxed{\Delta S' = -33.333 \frac{\text{J}}{\text{K}}} \quad +2$$

$$d) \bar{W}_{\text{net}} = \eta_{\text{th}} Q_{\text{in}} = \eta_{\text{th}} Q_H = (0.625)(26667) \quad +2$$

$$\boxed{\bar{W}_{\text{net}} = 16,666.875 \text{ J}} \quad +1$$

3) Assume: air is ideal $+1$

: $C_v, C_p = \text{const} \quad +1$

: quasi-st. $+1$

$$C_v = 720 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$R = 287 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$C_p = C_v + R = 1007 \frac{\text{J}}{\text{kg} \cdot \text{K}} \quad +1$$

$$a) \text{ Air: } \Delta S = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right) \quad +2$$

ideal,
 $C_v = \text{const.}$

$$\rightarrow 0 \quad T_2 = T_1$$

$$\Delta S = R \ln\left(\frac{V_2}{V_1}\right) = 287 \ln\left(\frac{4}{10}\right) \rightarrow \boxed{\Delta S = -262.98 \frac{\text{J}}{\text{kg} \cdot \text{K}}} \quad +1$$

$$b) \Delta S = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)$$

$$\rightarrow 0 \quad T_2 = T_1 \quad +1$$

$$= -287 \ln\left(\frac{1.7}{0.1}\right) \rightarrow \boxed{\Delta S = -813.13 \frac{\text{J}}{\text{kg} \cdot \text{K}}} \quad +1$$

$$c) \Delta S = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) \quad +1$$

$$\rightarrow 0 \quad P_2 = P_1$$

$$= 1007 \ln\left(\frac{1200}{300}\right) \rightarrow \boxed{\Delta S = 1396 \frac{\text{J}}{\text{kg} \cdot \text{K}}} \quad +1$$

d) Water is S.L.S. $+1 \rightarrow$ Table D.2

$$@ P = 1 \text{ MPa} \rightarrow \Delta S = S_g - S_f = 6.585 - 2.1381 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \quad +2$$

$$\boxed{\Delta S = 4.447 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}} \quad +2$$

4) Given: $\gamma = 1.4$

Assume: air is ideal gas +1

$C_p, C_v = \text{const} +1$

isotropic process +1

a) $P_1 = 0.1 \text{ MPa}$
 $V_1 = 0.2 \text{ m}^3/\text{kg}$
 $V_2 = 0.03 \text{ m}^3/\text{kg}$

$\rightarrow P_1 V_1^\gamma = P_2 V_2^\gamma +3$

$P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma = 0.1 \left(\frac{0.2}{0.03} \right)^{1.4}$

$\boxed{P_2 = 1.42 \text{ MPa}} +1$

b) $V_1 = 0.1 \text{ m}^3/\text{kg} \rightarrow$
 $T_1 = 400 \text{ K}$
 $T_2 = 500 \text{ K}$

$V_2 = V_1 \left(\frac{T_1}{T_2} \right)^{\frac{1}{\gamma-1}} +3$

$= (0.1) \left(\frac{400}{500} \right)^{\frac{1}{0.4}} \rightarrow \boxed{V_2 = 0.057 \frac{\text{m}^3}{\text{kg}}} +1$

c) $T_1 = 300 \text{ K} \rightarrow T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} +3$

$P_1 = 0.1 \text{ MPa}$

$P_2 = 1 \text{ MPa}$

$= 300 \left(\frac{1}{0.1} \right)^{\frac{0.4}{1.4}} \rightarrow \boxed{T_2 = 579.21 \text{ K}} +1$

d) $V_1 = 0.5 \text{ m}^3/\text{kg} \rightarrow V_2 = V_1 \left(\frac{P_1}{P_2} \right)^{\frac{1}{\gamma}} +1$

$P_1 = 5 \text{ MPa}$

$P_2 = 10 \text{ MPa}$

$= 0.5 \left(\frac{5}{10} \right)^{\frac{1}{1.4}} \rightarrow \boxed{V_2 = 0.305 \text{ m}^3/\text{kg}} +1$

e) $P_1 = 2 \text{ MPa} \rightarrow P_2 = P_1 \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} +3, 2$

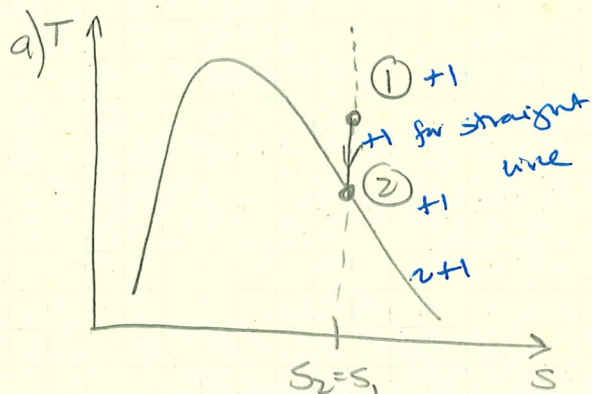
$T_1 = 400 \text{ K}$

$T_2 = 750 \text{ K}$

$= 2 \left(\frac{750}{400} \right)^{\frac{1.4}{0.4}} \rightarrow \boxed{P_2 = 18.05 \text{ MPa}} +1$

5) Given: $P_1 = 2 \text{ MPa}$
 $T_1 = 500 \text{ K}$
 $X_2 = 1$

Assume: S.C.S. +1
 : quasi-ef. +1
 : isentropic +1



State 1: $P_1 = 2 \text{ MPa}$

$T_1 = 500 \text{ K}$

$\Rightarrow$ Table D2: $T_1 > T_{\text{sat}}$
 : vapor

b) Vapor $\rightarrow$ NIST $\rightarrow$ $S_1 = 6.4265 \text{ kJ/kg-K}$ +2

c) $S_2 = S_1 = 6.4265 \text{ kJ/kg-K}$ +2

d) Interpolate between:

$\rightarrow T = 471.44 \text{ K}$ $P = 1.5 \text{ MPa}$ $S_g = 6.4430 \text{ kJ/kg-K}$ +1/2

$\rightarrow T = 474.52 \text{ K}$ $P = 1.6 \text{ MPa}$ $S_g = 6.4199 \text{ kJ/kg-K}$ +1/2

$\Rightarrow$ $T_2 = 473.64 \text{ K}$ +1
 $P_2 = 1.57 \text{ MPa}$ +1

6) Given: $\frac{V_1}{V_2} = 11.2$
 $\gamma = 1.4$

Assume: Ideal gas +1
 const c_p, c_v +1
 isentropic +1
 quasi-ef. +1

a) $P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma$ +3 +2 for correct plot

b) $T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$ +3 +2 for correct plot

c) any reasonable answer +6